



LC Mechanics EXAM Paper for Revision

LC Mechanics (University of Birmingham)



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*Calculators may be used in this examination but must not be used to store text.
Calculators with the ability to store text should have their memories deleted prior
to the start of the examination.*

Special Requirements: Multiple Choice Answer Sheets

UNIVERSITY OF BIRMINGHAM

School of Engineering

Level C

04 28608

Mechanics 1

Sample Examination for Revision

Time Allowed: 2 hours

Answer **ALL** questions, by marking the appropriate answer in pencil on the Multiple Choice Answer Sheet provided. Any changes must be made by deleting the incorrect answer with an eraser.

All questions carry equal marks.

In these questions $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, are three mutually perpendicular vectors that define a right-handed Cartesian coordinate system.

The magnitude of the acceleration due to gravity is given by $g = 9.81 \text{ m.s}^{-2}$.

1. A lift (mass, m) accelerating upwards with an acceleration of magnitude a . The lift is suspended by a steel cable. The direction of acceleration is defined to be positive. This acceleration is in the opposite direction to gravity (magnitude, g). Which ONE of the following is equal to the tension in the cable:
 - a) the weight of the lift
 - b) ma
 - c) $m(g-a)$
 - d) $m(a-g)$
 - e) $m(a+g)$

2. A person walks from the centre of a boat to the edge of the boat. Neglecting viscosity and friction, which ONE statement about the movement of the boat relative to a fixed external point (e.g. the nearest shore), as the result of the person moving, is FALSE?
 - a) the boat moves because its centre of gravity moves
 - b) the boat moves because the person's feet exert a force on it
 - c) if the boat and person are considered to be a system, then there is no net external force on the system
 - d) if the boat and person are considered to be a system, then the centre of gravity of the system moves as the person moves
 - e) if the boat and person are considered to be a system, then the centre of gravity of the system does not move as the person moves.

3. A rectangular cuboid is sheared by a force, F , applied to its top surface (shown by the arrow in the figure, below) while its base is fixed in place. The cuboid has a length L , height H , and width W . The shear stress experienced by the cuboid is

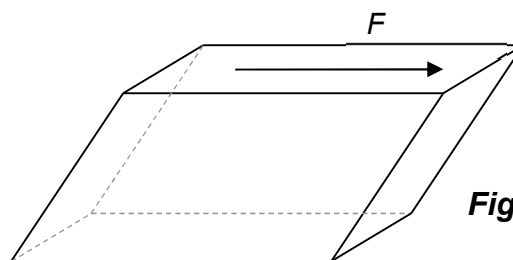
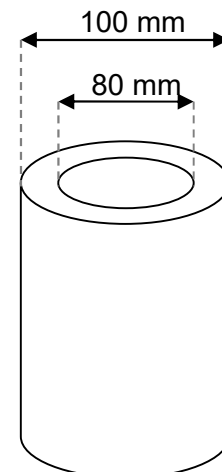


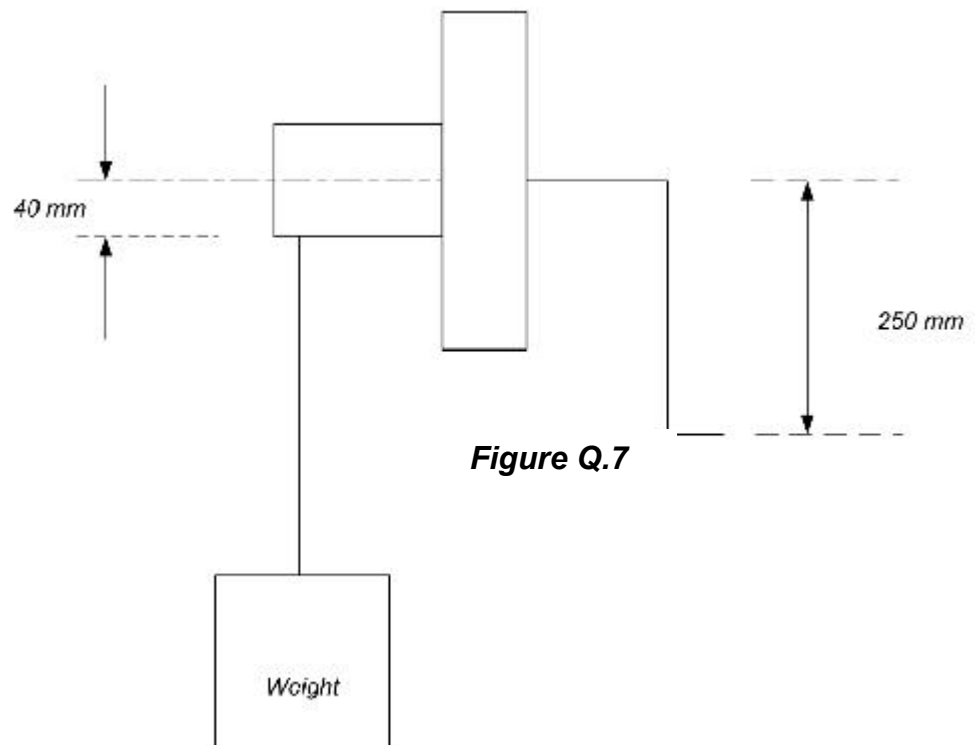
Figure Q.4

- a) F / LH
- b) F / LW
- c) F / HW
- d) $\frac{1}{2} F / HW$
- e) not calculable without the shear modulus.

4. The force $\mathbf{F} = 3.6\mathbf{i} + 1.7\mathbf{j} + 1.8\mathbf{k}$ N acts at $\mathbf{r} = 2.8\mathbf{i} + 1.3\mathbf{j} + 4.2\mathbf{k}$ m, relative to a point P. The rate of change of angular momentum exerted about P is:
- $-4.8\mathbf{i} + 10.1\mathbf{j} + 0.1\mathbf{k}$ N.m
 - $-4.8\mathbf{i} - 10.1\mathbf{j} + 0.1\mathbf{k}$ N.m
 - $4.8\mathbf{i} - 10.1\mathbf{j} - 0.1\mathbf{k}$ N.m
 - $4.8\mathbf{i} + 10.1\mathbf{j} - 0.1\mathbf{k}$ N.m
 - 5.4 N.m.
5. If G is the universal gravitational constant, the speed, v , required to launch a space vehicle of mass m so that it escapes the gravitational attraction of a planet of mass M and radius R is:
- $\sqrt{\frac{2GM}{R}}$
 - $\sqrt{\frac{2Gm}{R}}$
 - $\sqrt{\frac{GM}{R}}$
 - $\sqrt{\frac{2Gm}{R^2}}$
 - $\sqrt{\frac{GM}{R^2}}$.
6. Which ONE of the following is the CORRECT normal stress induced through the cross section of the hollow cylinder in **Figure Q.6**, when a force of 130 kN acts on its surface:
- 414 MPa
 - 103 kPa
 - 46 MPa
 - 11 MPa
 - 12 kPa.

Figure Q.6

7. A winch (**Figure Q.7**) with a handle 250 mm long and radius of 40 mm is used to support a weight of 3300 N via a light, inextensible string. Turning the handle moves the weight along a straight line. Frictional losses in the winch result in it being only 40% efficient. The mechanical advantage of the system is:
- 16
 - 10
 - 6.3
 - 3.8
 - 2.5.



8. A rectangular beam, 0.60 m in height, h , and 0.15 m in width, is subjected to a bending moment, M , about its neutral axis of 75.0 kN.m. Noting that stress, $\sigma = -\frac{My}{I}$ and $I = \int_A y^2 dA$ where A is the cross-sectional area, and y the position along a y -axis perpendicular to the length of the beam, with its origin at the neutral axis. The maximum bending stress induced is:
- 8.3 MPa
 - 4.2 MPa
 - 2.1 MPa
 - 17 kPa
 - 9.0 kPa

9. The rotational inertia of a uniform and homogenous thin rod, of mass m and length L , rotating about an axis perpendicular to the length of the rod (**Figure Q.9**) that passes through its centre of gravity is:

- a) mL^2
- b) $\frac{1}{3}mL^2$
- c) $\frac{1}{12}mL^2$
- d) $\frac{1}{3}mL^3$
- e) $\frac{1}{12}mL^3$.

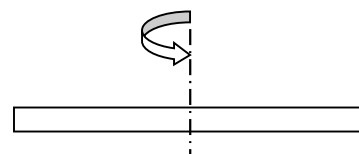


Figure Q.9

10. A ledge is supporting a 3 kg load, which has its centre of mass at its geometric centre ($Cg1$, **Figure Q.10**). The ledge itself has a weight of 1.1 kg (its centre of gravity is at $Cg2$ in **Figure Q.10**). A light, inextensible, string connects the ledge to a wall under tension. The ledge is also supported by a pin connected to the ground. Which ONE of the following statements is FALSE for the system to be in equilibrium:

- a) the tension in the string is 180 N
- b) the reaction force acting along the x-axis through the centre of the pin has a magnitude of 62 N
- c) the reaction force acting along the y-axis through the centre of the pin has a magnitude of 140 N
- d) the torque about the centre of the pin, due solely to the string, has a magnitude of 11 N.m
- e) both the sum of the forces and torques acting on the system is 0.

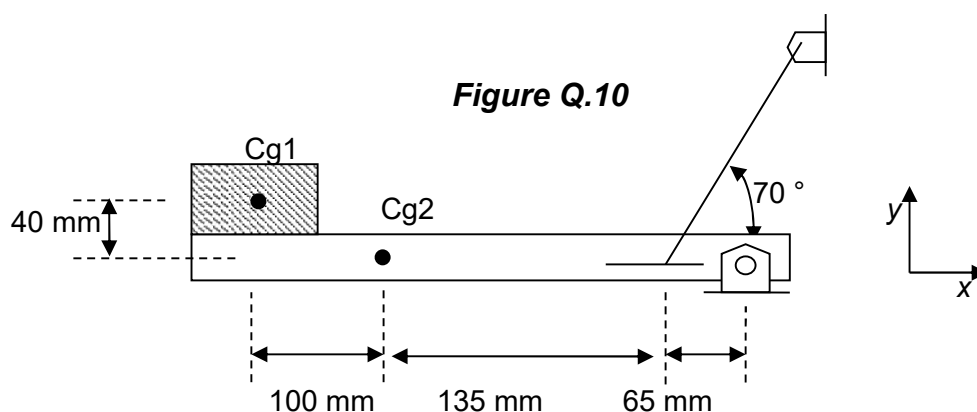


Figure Q.10

11. The cantilever beam shown in **Figure Q.11** is subjected to a 3 kN-m couple. Knowing that $E = 165 \text{ GPa}$ and neglecting the effect of fillets, what is the maximum compressive stress?

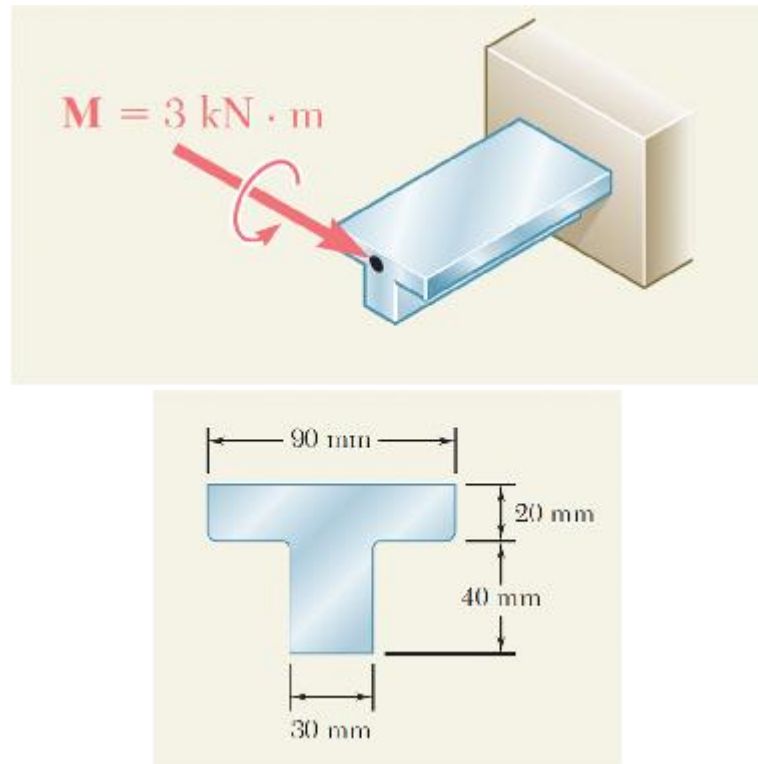


Figure Q.11

- a) -30 MPa
 b) -76.0 MPa
 c) -131.3 MPa
 d) -518.2 MPa
 e) -207.4 MPa
12. In the above question, what is the radius of curvature of the cantilever beam?
- a) 20.9 m
 b) 82.6 m
 c) 12.1 m
 d) 70.2 m
 e) 47.7 m

13. The beam shown in **Figure Q.13** is made of aluminium and has a uniform 3-mm wall thickness. If the shear force in the beam is 5 kN, what is the shearing stress at point A? (Note: The dimensions given are to lines midway between the outer and inner surfaces of the beam)

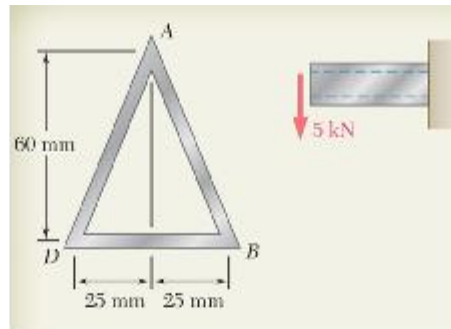


Figure Q.13

- a) 5 MPa
 b) 2.5 MPa
 c) 9.3 MPa
 d) 4.6 MPa
 e) 0 MPa
14. In the above question, what is the maximum shear stress in the beam?

- a) 2.5 MPa
 b) 4.6 MPa
 c) 18.54 MPa
 d) 9.3 MPa
 e) 5 MPa

15. Determine the magnitude of the vertical support reaction at D for the structure shown in **Figure Q.15**.

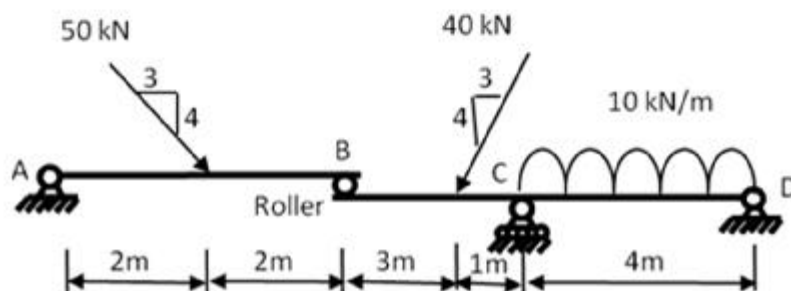


Figure Q.15

- a) 8 kN
 b) 30 kN
 c) 0 kN
 d) 24 N
 e) 20 N.

16. Calculate the 2nd moment area of the shape in **Figure Q.16** about its centroid.

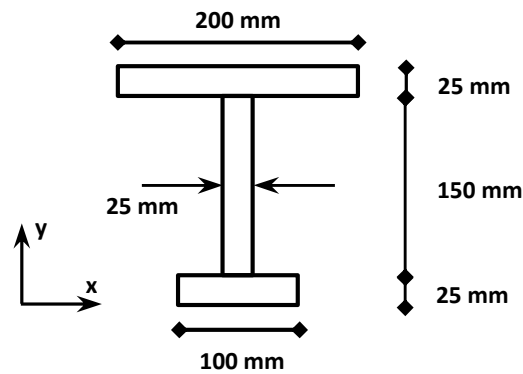


Figure Q.16

- a) $60 \times 10^6 \text{ mm}^4$
- b) $7.4 \times 10^{-6} \text{ mm}^4$
- c) $50 \times 10^{-6} \text{ mm}^4$
- d) $26 \times 10^6 \text{ mm}^4$
- e) $13 \times 10^6 \text{ mm}^4$

17. The cantilever beam AB is of uniform cross section and carries a load P at its free end A. Determine the deflection at A.

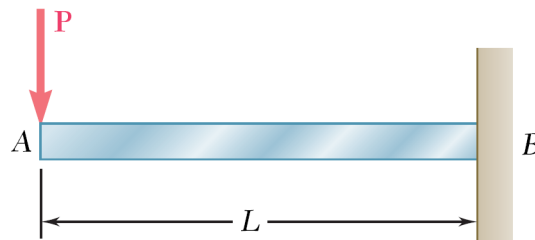


Figure Q.17

- a) $PL^3/(3EI)$
- b) $PL/(EI)$
- c) $PL/(2EI)$
- d) $PL^2/(EI)$
- e) $PL^2/(2EI)$.

18.

Each of the four vertical links connecting the two rigid horizontal members is made of aluminium ($E = 70 \text{ GPa}$) and has a uniform rectangular cross section of $10 \times 40 \text{ mm}$. For the loading shown, what is the deflection of point G (in μm).

- a) $\delta_G = 390 \mu\text{m} \downarrow$
- b) $\delta_G = 209 \mu\text{m} \downarrow$
- c) $\delta_G = 80 \mu\text{m} \downarrow$
- d) $\delta_G = 104 \mu\text{m} \downarrow$
- e) $\delta_G = 150 \mu\text{m} \downarrow$

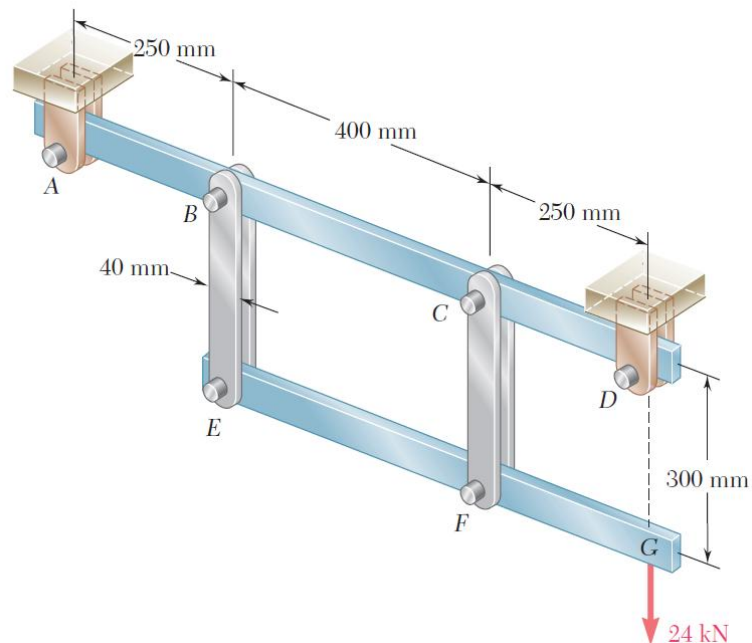
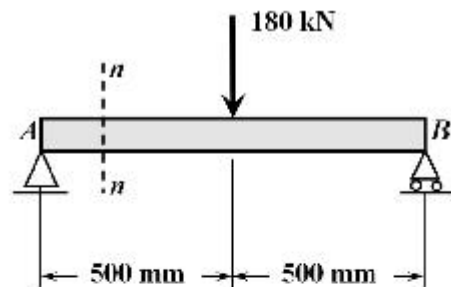


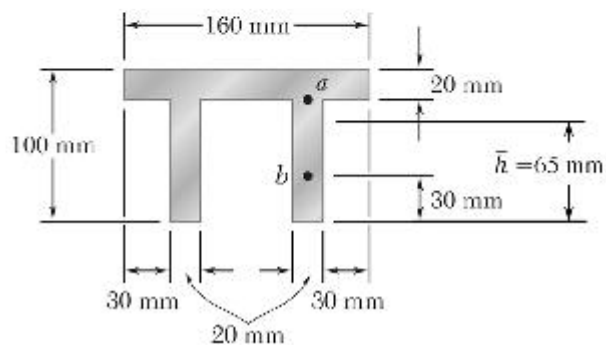
Figure Q.18

19. For the beam and loading shown in **Figure Q.19**, considering *section n-n*, what is the shearing stress at point *a*. The location of the centroid of the section is given in the **Figure Q.19b** and the centroidal moment of inertia is $5.8133 \times 10^6 \text{ mm}^4$.

- a) $\tau = 31 \text{ MPa}$
- b) $\tau = 23 \text{ MPa}$
- c) $\tau = 40 \text{ MPa}$
- d) $\tau = 45 \text{ MPa}$
- e) $\tau = 0 \text{ MPa}$



(a)



(b)

Figure Q.19

20. For the beam and loading shown in **Figure Q.19**, what is the maximum shear stress in *section n-n*.

- a) $\tau = 30 \text{ MPa}$
- b) $\tau = 20 \text{ MPa}$
- c) $\tau = 40 \text{ MPa}$
- d) $\tau = 45 \text{ MPa}$
- e) $\tau = 33 \text{ MPa}$